POST-MORTEM · ARGOSY DESKTOP

Project post-mortem

Interim review after Phase 4 (Phase 5 in progress): claude.ai planning, Claude Code development

PROJECT	Argosy Desktop
PERIOD	2026-10-16 → 2026-11-06
SURFACES	claude-ai, claude-code
FINDINGS	9 confirmed · 10 plausible · 2 rejected
RECOMMENDATIONS	12 P1 · 12 P2 · 5 P3
SIGNALS REVIEWED	3
REVISION	r1 · generated 2026-11-07
STANDARD	POST-MORTEM-STANDARD v1

1. Executive summary

Delivery speed was never the constraint. Phases 0-4, planned as 28 weeks, closed in 9 calendar days on 2026-10-24, with a heavily scripted restructure, a verification gate at every step and unusually candid "not verified" notes. What limited the project was everything around the code: usage, owner-held decisions and checks, and test isolation. On day 2 the development context reached about 455k tokens and work stopped for about 90 hours under the owner's weekly usage threshold. Phase 5 has since idled for most of a week on four owner items, the Game Mode fix is still unverified on real hardware, and the fork has fallen 13 stable upstream releases behind.

The main causes are process defaults that arrived late or not at all. Work units, handoffs and pacing were adopted as a fix on day 2 rather than as the starting setup. Desktop UI walks ran in the owner's live session until the Phase 5 VM lab, which sent the owner's installed-app list to Google's Play Store lookup and changed real settings. Owner questions and hardware checks were recorded in files and batched at step or phase ends, with no alert. No cadence was ever set for merging upstream releases. The planning session added its own errors: the first plan borrowed another project's design, and its week-based schedule never matched agent-driven delivery.

What to change: make GUI tests refuse to run against the owner's real session; merge each upstream release within a week and report the lag daily; push a phone notification when the pacer waits on the owner; attach a short hardware checklist to every alpha; and fold the unit/handoff/pacing scheme into the Token Usage Standard as the default for multi-day projects.

Top recommendations

PRI	ID	ACTION	TARGET	SCORE	DECISION
P1	PM-010-R2	Decide on the upstream proposal now: post it, or record 'long-term fork' and plan for the merge cost.	user	14.1	proposed
P1	PM-007-R2	Add 'GUI tests never run against the owner's live session or data; scripts must check' as a required rule in the VM Lab Standard.	standard:VM-LAB-STANDARD	12.8	proposed
P1	PM-010-R1	Merge each upstream stable release into main within a week using reapply.sh (at least at every phase gate), and add 'upstream releases behind' to the daily update.	claude-md	8.8	proposed
P1	PM-001-R1	When a request matches more than one remembered project, name the project and the scope the plan assumes in the first reply, and ask if the request does not settle it, before drafting.	workflow	8.1	proposed
P1	PM-007-R1	Run every GUI walk in the VM lab or a disposable user with separate HOME/XDG dirs and testbed-only network, and make the walk scripts refuse to start against the owner's real HOME.	project	8	proposed
P1	PM-003-R1	Run build-heavy projects in Claude Code on the owner's machine (or the VM lab) from the first execution session, and keep claude.ai/Cowork for planning and reviews.	workflow	7.7	proposed

Score = severity × benefit × confidence ÷ effort (P1 ≥ 5, P2 ≥ 2, else P3). Full register in §5.

2. Project at a glance

1	12.3M	106	3.8%
CLAUDE CODE SESSIONS	TOKENS (ALL KINDS)	TOOL CALLS	TOOL ERROR RATE

MEASURE	VALUE
Transcript span	2026-10-16 → 2026-10-16
Subagent sessions	0
Tokens: input / output	76 / 39,999
Tokens: cache read / write	11,999,071 / 285,129
User messages / interrupts	4 / 1
Git	224 commits · 0 reverts · 1 fix-type commits · 2026-10-17 → 2026-10-24
Plan	no docs/PLAN.md
Journal	missing

Tool use

TOOL	CALLS	ERRORS	ERROR %
Bash	27	1	3.7
mcp__Claude_Docs__update	23	2	8.7
mcp__Claude_Docs__read	8	1	12.5
WebSearch	7	0	0
ToolSearch	5	0	0
TaskCreate	5	0	0
Write	5	0	0
mcp__Claude_Docs__guide	4	0	0
mcp__Claude_Docs__batch	4	0	0
TaskUpdate	3	0	0
mcp__memory__memory_read	2	0	0
WebFetch	2	0	0

Most-changed files (git)

FILE	LINES CHANGED
scripts/desktop/restructure/moves.txt	1,841
DEVLOG.md	1,039
scripts/desktop/restructure/slices.py	992
shared/src/desktopMain/kotlin/com/nendo/argosy/data/storage/StorageAttributionRepository.desktop.kt	979
scripts/desktop/restructure/movable.py	799
scripts/desktop/restructure/rewrite_imports.py	614
docs/desktop/handoff/phase-3.md	575

shared/src/desktopMain/kotlin/com/nendo/argosy/data/emulator/PlaySessionTracker.desktop.kt	569

Timeline

DATE	SURFACE	EVENT
2026-10-16	claude-ai	Plan drafted; first version used another project's web-UI/Electron design; owner redirected to a faithful port; rewritten around Compose Multiplatform
2026-10-16	claude-ai	Execution session: package hosts blocked in the cloud; survey and handoff written; owner moved development to Claude Code
2026-10-16	claude-code	Toolchain installed, three spikes pass; Phase 0 closed. Phase 1 (planned weeks 3-8) closed the same day; upstream v2.24.0 merged
2026-10-17	claude-code	Phase 2 seam batches; context ~455k tokens; owner adds token-economy and pacing rules; paused for the weekly usage reset
2026-10-21	claude-code	Resumed after ~90 h; testbed containers recreated
2026-10-22	claude-code	Desktop shell, input and all reachable screens in one day; Phase 2 closed with carried-forward items; Phase 3 started on the owner's go-ahead
2026-10-23	claude-code	Phase 3: registry, host launch, save sync, alpha Flatpak on GitHub Pages; walk incidents on the owner's desktop; owner Bazzite test 15/3/13; four plan changes; Phase 3 closed
2026-10-24	claude-code	Phase 4 (re-scoped as Linux adaptation, 9 steps) closed; alpha published at every step; Phase 5 steps 1-3 closed
2026-10-27	claude-code	Owner re-frames goal as Linux adaptation, not a 1:1 clone; side-by-side step dropped; step 4 reopened; VM lab approved
2026-10-28	claude-code	VM lab built; owner pauses mid-build (two VMs rebuilt); resumed 10-29
2026-10-30	claude-code	Startup abort diagnosed and fixed; Flathub groundwork done; remaining work needs the owner
2026-11-04	claude-code	Owner requests: exit row moved, hover highlight, resets; exit cleanup found dead since 10-24
2026-11-05	claude-code	VM tests on GNOME and gamescope; pacer waiting-owner on four items
2026-11-06	upstream	Upstream at v2.29.1, 13 stable releases ahead of the fork's v2.24.0 base

3. Method

1. **Evidence.** Claude Code transcripts, git history, the plan, the usage and lab records, and the session journal (the only record of Cowork sessions) were mined with `postmortem.mjs mine` into 3 machine signals.
2. **Analysis.** The work was reviewed through nine lenses: Process efficiency, Logic & correctness, Planning & scope, Token & usage, Tooling & automation, Environment & lab, Communication & handoffs, Quality & testing, Standards & ai-assets.
3. **Audit.** Every observation was handed, with its evidence but not its conclusion, to an auditor who tried to disprove it. Verdicts: **confirmed** (evidence holds, confidence ≥ 70), **plausible** (supported but not proven), **rejected** (disproved or unsupported; listed in Appendix A). Self-audits are capped at confidence 70.
4. **Evaluation.** Every recommendation states benefit and effort (1–5), risks, the alternatives considered and why this option was chosen. Score = severity × benefit × (confidence ÷ 100) ÷ effort.
5. **Completeness.** Every machine signal is tied to a finding or explicitly dismissed with a reason (Appendix B). The report cannot be generated otherwise.

Notes and limits

Interim review: Phase 5 is in progress (pacer waiting-owner since 2026-11-05). Run from claude.ai (Cowork-style surface) linked to the owner's computer. The Claude Code transcripts in ~/.claude on the owner's machine are a protected location and could not be read, so there are no token, tool-call or interrupt metrics for the Claude Code sessions; the only mined transcript is the 2026-10-16 claude.ai planning session (truncated before this review began). Git metrics come from the GitHub copy of main (through 2026-10-24, since the push rule allows pushes only at a phase close); the 27 unpushed Phase 5 commits through 2026-11-05 were read from the local git reflog. The git churn figures include upstream's own commits merged on 2026-10-16. The plan lives in PORT_PLAN.md, which the miner does not parse (it reads docs/PLAN.md), so plan progress was taken from PORT_PLAN.md, the handoffs and DEVLOG by hand. There is no session journal; Claude Code sessions were reconstructed from the 410 KB DEVLOG (read in four parts by subagents), the per-phase handoffs, gotchas.md and the reflog. Each observation was audited by an independent reviewer subagent (opus for severity 4 and logic findings, sonnet otherwise) given the claim and evidence references only; reviewers corrected numbers in 11 observations and rejected two.

4. Findings

Process efficiency

PM-014 **Test-environment overhead paid again and again** PLAUSIBLE severity 2/5 · confidence 45%

OBSERVATION

Environment overhead recurred: a save proof's snapshot restore forced a desktop re-pair; the wizard walk recipe needed fixing before walks passed; in Phase 5 the VM lab took several units over 10-28 to 10-30 and more fixes to 11-05, two VMs were rebuilt after a mid-build pause, Ubuntu runs hung, and the log says 'every failure along the way was the lab, not the app'. Walks use fixed sleeps that drifted, and gotchas records two self-kill hazards (pkill -f, pgrep -f).

IMPACT

Several Phase 5 units spent on lab fixes rather than the product; walk-recipe rework in Phases 3-4.

EVIDENCE

1. FILE DEVLOG.md:888-892 — snapshot restore invalidated the pairing; re-pair needed
2. FILE DEVLOG.md:1509-1537,1633 — VM lab build issues; 'every failure … was the lab, not the app'
3. FILE docs/desktop/gotchas.md:6,70 — pkill -f and pgrep -f self-match hazards

ROOT CAUSE

Test infrastructure was built inside product phases, under time pressure, with no reusable, validated base images or readiness signals.

> AUDIT · REVIEWER SUBAGENT (SONNET) (INDEPENDENT)
>
> **Method:** Read D:886-902, 958, 1020, 1174-1191, 1509-1537, 1567, 1575, 1627-1634 and gotchas G:6/56/58/62/68/70/76; grepped for re-pair, pkill and pgrep. Re-scoped to severity 2 and narrowed the wording.
>
> **Counter-evidence sought:** Only one re-pair is documented (D:892, plus a mention at D:958), not one per proof; no count of three wizard failures at D:1176-1190; lab units ran 10-28 to 10-30 (about 3-4 days), not a week; gotchas hold two self-kill notes (G:6 pkill -f, G:70 pgrep -f), not three; the VM rebuild followed an owner-initiated pause. Fixed-sleep drift (D:1567), repeated lab-script fixes and 'every failure … was the lab' (D:1633) hold.

PM-014-R1 P3 **Build validated golden VM images per distro once (gamescope and testbed pairing baked in) and replace fixed sleeps with readiness markers from the app log.**

TARGET	standard:VM-LAB-STANDARD
BENEFIT / EFFORT	3/5 · 3/5 → score 0.9
RISK	Up-front effort; images need refreshing when distros update.
ALTERNATIVES	Keep fixing the lab as issues appear.
EVALUATION	Pays back across every later walk and every future project that uses the lab.
DECISION	proposed

PM-014-R2 P3 **Make save-proof.sh keep or restore the desktop's pairing so a proof never needs a separate re-pair.**

TARGET	project
BENEFIT / EFFORT	2/5 · 1/5 → score 1.8
RISK	Touches the save-proof script, which guards a fragile zone.
ALTERNATIVES	Keep running desktop-repair.sh after each proof.
EVALUATION	Removes a manual step repeated at every save-zone change.
DECISION	proposed

Logic & correctness

PM-011 **Silent desktop stubs recurred as a class** PLAUSIBLE severity 3/5 · confidence 70%

OBSERVATION

Desktop stand-ins that silently misbehaved recurred across Phase 2: 18 TODO() initializers threw at startup, emptyFlow() stubs blocked combine() twice (p4h, then p4m, after a value-flow sweep missed one member), stubs returning false hid storage and media paths, and a QuayPass stub was hard-coded DISABLED. The TODO and value-flow cases got class sweeps; the constant-returning stubs were found by walks one at a time, with no audit of them.

IMPACT

Several walk-and-diagnose cycles in Phase 2 (blank screens, endless spinners, wrong labels).

EVIDENCE

1. FILE `DEVLOG.md:704,739` — emptyFlow blocked combine in p4h and again in p4m
2. FILE `DEVLOG.md:652,731,755,763` — TODO initializers, isPathAvailable/isGranted false, QuayPass DISABLED
3. FILE `docs/desktop/gotchas.md:13` — TODO in a desktop initializer throws at injection

ROOT CAUSE

Placeholders were written to make the desktop compile, with no rule or check that a stand-in must either behave like the real thing or fail loudly.

> AUDIT · REVIEWER SUBAGENT (OPUS) (INDEPENDENT)
>
> **Method:** Read DEVLOG 652, 703-704, 731-733, 739-741, 755, 763 and gotchas line 13. Re-scoped: class sweeps were done but narrowly scoped; no audit of constant-returning stubs.
>
> **Counter-evidence sought:** Two fixes were class-level sweeps (all 18 TODO initializers at once; every value-flow stub converted to flowOf at p4h with a stated reason not to ban emptyFlow), so 'case by case' overstates it. The second emptyFlow hit came from misclassifying packageChanges as an event stream. The false/DISABLED stubs were each found by walks and fixed singly, with no recorded audit of constant-returning stubs.

PM-011-R1 P2 **Add a gate check that fails on emptyFlow(), TODO() and constant false/null actuals in desktopMain unless listed in an allowlist with a reason.**

TARGET	`tool:stub-guard`
BENEFIT / EFFORT	3/5 · 2/5 → score 3.2
RISK	False positives on legitimate constants; allowlist maintenance.
ALTERNATIVES	Code review only; rely on walks.
EVALUATION	A grep-level check is cheap and catches the class before a walk does.
DECISION	proposed

Planning & scope

PM-005 **Week-based schedule never matched agentic delivery** CONFIRMED severity 2/5 · confidence 85%

OBSERVATION

The plan's 32-week schedule (one developer at ~20 h/week with AI-assisted coding, a per-hour estimate written in the planning session) bore no relation to delivery: Phases 0-4, planned for weeks 1-28, closed within 9 calendar days (about 4 of them idle), and Phase 1 (weeks 3-8) closed on its first day. The real constraints were usage windows and owner availability. The roadmap was never re-baselined and still says about 32 weeks.

IMPACT

Schedule and progress figures carried no information; nobody could see the real limits (usage, owner time) in the plan.

EVIDENCE

1. FILE PORT_PLAN.md Roadmap — 'Six phases over about 32 weeks'; status line says Phases 0-4 complete
2. FILE docs/desktop/handoff/README.md — Phase 0 and 1 closed 2026-10-16; Phase 4 closed 2026-10-24
3. LOG git reflog — local commits 2026-10-16 .. 2026-11-05

ROOT CAUSE

The estimate model counted developer hours; agent-driven throughput is bounded by usage windows and owner decisions, which the plan did not model.

> AUDIT · REVIEWER SUBAGENT (OPUS) (INDEPENDENT)
>
> **Method:** Read PORT_PLAN.md line 283 and roadmap table, handoff README close dates, reflog (Phase 1 approved 12:25 and closed 21:40 on 10-16; Phase 4 close 10-24 14:11), DEVLOG 500 and 542, pacer json.
>
> **Counter-evidence sought:** Checked whether the roadmap was ever re-baselined: only a scope note was added, the weeks never changed. The estimate assumed AI-assisted coding at 20 h/week, so 'human-developer estimate' is imprecise (a per-hour estimate). The 9 days include about 4 idle days (10-18 to 10-20). The first (Electron) plan said about 7 months.

PM-005-R1 P2 **Replace the week-based roadmap with actual dates plus remaining work in units, and re-baseline at every phase gate, naming the binding constraint (usage window or owner action).**

TARGET	project
BENEFIT / EFFORT	2/5 · 1/5 → score 3.4
RISK	Estimates in units are less familiar to readers outside the project.
ALTERNATIVES	Keep weeks and add a note; drop estimates altogether.
EVALUATION	Keeps a forecast that means something; dropping estimates removes the owner's planning aid.
DECISION	proposed

PM-006 Which features to keep was decided after porting PLAUSIBLE severity 3/5 · confidence 70%

OBSERVATION

The approved plan, following the owner's instruction to preserve as much of the feature set as possible, marked every feature Port or Adapt except two deferrals. After a first look at the alpha (10-23) the owner removed Steam, Jellyfin, Social and the Android Settings row and dropped built-in emulation, netplay and dual-screen (one of four plan changes that day; the other three added desktop features), and hid Check-In, which Phase 2 had ported and walked. On 10-27 the goal was re-framed as Linux adaptation, not a 1:1 clone, and an Android side-by-side step was dropped after four units of work.

IMPACT

Screens ported and walked in Phase 2 later hidden; about four units of side-by-side tooling deleted; Phase 4 re-planned mid-project.

EVIDENCE

1. FILE DEVLOG.md:1073-1096 — four 'Plan change (owner decision)' entries on 2026-10-23
2. FILE DEVLOG.md:1495-1503 — 'not a 1:1 clone'; side-by-side step dropped, checklist and side-by-side.sh deleted
3. FILE PORT_PLAN.md Desktop divergences — removed and not-in-this-version tables

ROOT CAUSE

Requirements changed once the owner used the product: the plan encoded 'preserve everything' without a row-by-row keep/drop review, and the owner first saw a usable build only at the Phase 3 alpha.

> **AUDIT · REVIEWER SUBAGENT (OPUS) (INDEPENDENT)**
>
> **Method:** Read DEVLOG 1073-1096, 1179-1184, 751-757, 1353-1402, 1495-1503, the original PORT_PLAN.md in commit 02ee5cfc8 and transcript idx 44 and 57. Corrected several details.
>
> **Counter-evidence sought:** The approved plan had two Defer rows (QuayPass, camera), so not 'every feature Port or Adapt'; only one of the four 10-23 plan changes removed features (three added desktop features); Check-In was hidden by a separate decision the same day. The owner's planning instruction was to preserve as much of the feature set as possible, so the 'nothing dropped' plan followed it rather than being an oversight; the scope change reflects requirements that emerged in use.

PM-006-R1 P2 **At plan approval, walk the parity matrix row by row with the owner as keep / remove / later, and put a clickable build in front of the owner by the end of the first build phase.**

TARGET	workflow
BENEFIT / EFFORT	3/5 · 2/5 → score 3.2
RISK	Earlier decisions may still change once the owner uses the product.
ALTERNATIVES	Port everything and prune later (as happened); ask only at phase gates.
EVALUATION	A row-by-row pass surfaces most keep/drop choices while they are cheap; an earlier clickable build covers the ones that only appear in use.
DECISION	proposed

PM-010 **Fork has fallen 13 upstream releases behind** CONFIRMED severity 4/5 · confidence 88%

OBSERVATION

The plan's top risk has materialised. main was last merged with upstream at v2.24.0 (10-16) while upstream is at v2.29.1 (11-06): 13 stable releases plus a beta, 319 commits, 987 files changed since the merge base. The re-apply script was rehearsed and used once; the plan intended re-applying each release but no cadence or trigger exists, and the upstream proposal to share one codebase is still unposted.

IMPACT

Desktop app behind Android by 13 releases; merge cost and conflict risk grow with every release; the plan's core 'identical to Android' promise erodes.

EVIDENCE

1. COMMIT 4508ad68d — only upstream merge: v2.24.0, 2026-10-16
2. LOG `git log 98562a1..origin/upstream` — 319 commits, 14 version bumps 2.25.0-2.29.1 (one beta), 987 files
3. FILE DEVLOG.md:76-77 — proposal held; 'posting stays with the owner'

ROOT CAUSE

The workflow syncs the read-only upstream branch automatically but nothing schedules merging it into main, and the proposal decision has no deadline.

> AUDIT · REVIEWER SUBAGENT (OPUS) (INDEPENDENT)
>
> **Method:** Ran git merge-base origin/main origin/upstream (98562a1e7, v2.24.0), git log base..origin/upstream (319 commits), git diff --shortstat (987 files), counted version bumps; checked reflog and git log --merges for later merges; read DEVLOG 76-77, 341, 367-368, PORT_PLAN risks, CLAUDE.md.
>
> **Counter-evidence sought:** Exactly one upstream merge (4508ad68d) through 11-05, none in unpushed local commits. Of the 14 bumps one is 2.27.0-beta.1, so 13 stable releases plus a beta. reapply.sh was rehearsed once and used once for the real merge. The plan intended re-applying each release (PORT_PLAN risks) but no schedule or trigger exists. No record of the proposal being posted.

PM-010-R1 P1 **Merge each upstream stable release into main within a week using reapply.sh (at least at every phase gate), and add 'upstream releases behind' to the daily update.**

TARGET	claude-md
BENEFIT / EFFORT	5/5 · 2/5 → score 8.8
RISK	Each merge needs the Android gate, A/B and save proof; conflicts may need owner calls.
ALTERNATIVES	Merge once before 1.0; freeze on v2.24.0.
EVALUATION	Small regular merges keep conflicts small; one big merge before 1.0 risks a long stall.
DECISION	proposed

PM-010-R2 P1 **Decide on the upstream proposal now: post it, or record 'long-term fork' and plan for the merge cost.**

TARGET	user
BENEFIT / EFFORT	4/5 · 1/5 → score 14.1
RISK	Upstream may decline or ask for changes.
ALTERNATIVES	Keep holding it.
EVALUATION	Either answer removes uncertainty; holding it keeps the most expensive option open by default.
DECISION	proposed

Token & usage

PM-004 **Unbounded context spent the usage budget on day 2** PLAUSIBLE severity 4/5 · confidence 75%

OBSERVATION

By 2026-10-17 the development session held about 4,400 messages and a ~455k-token context, with an orchestrating session of ~370k woken every turn. The owner imposed a token-economy rule (work units, handoff file, quiet build output) and usage pacing mid-phase, then, having reached the owner's own weekly threshold (lowered to 50%), paused development until the weekly reset: no commits from 10-17 21:27 to 10-21 15:29, and the testbed containers had vanished meanwhile.

IMPACT

About 90 hours of stalled calendar time, a weekly usage window largely spent, and testbed rebuild on resume.

EVIDENCE

1. FILE DEVLOG.md:500-511 — pacing rule; 'Measured cost driver: context size, not the work'; ~455k-token context
2. FILE DEVLOG.md:542-548 — paused until weekly reset; resumed 10-21; testbed containers had disappeared
3. LOG `git reflog` — 90 h gap 2026-10-17 21:27 → 2026-10-21 15:29

ROOT CAUSE

Development ran as unbounded conversations with build output in context; unit boundaries, handoffs and pacing were introduced as a fix after the cost appeared, not as the starting setup.

> AUDIT · REVIEWER SUBAGENT (OPUS) (INDEPENDENT)
>
> **Method:** Read DEVLOG 500-511 and 542-550, reflog 10-17 to 10-21, CLAUDE.md token-economy rules, and the 50% cap commit a343c70f3. Re-scoped: two large contexts; the stop came from the owner's own weekly threshold.
>
> **Counter-evidence sought:** Context figures confirmed (development session ~4,400 messages / 455k tokens, plus an orchestrating session ~370k woken every turn, D:504), rules imposed mid-Phase 2, 90.0 h commit gap and vanished containers (D:548) confirmed. Nothing shows the platform's weekly limit being hit: the last commit before the gap lowers the owner's own weekly threshold to 50%, and the pause was until the weekly reset. Work may have stopped before 21:27 (last product commit 19:29).

PM-004-R1 P1 **Make work units, a rewritten per-phase handoff, quiet build output and usage pacing the default bootstrap of every multi-day project (fold this project's scheme into the Token Usage Standard).**

TARGET	`standard:TOKEN-USAGE-STANDARD`
BENEFIT / EFFORT	4/5 · 2/5 → score 6
RISK	More ceremony per unit; risk of over-fragmenting short projects (apply only to multi-day work).
ALTERNATIVES	Rely on auto-compaction; ask the owner to watch usage.
EVALUATION	The scheme demonstrably held from 10-17 to the end without a second weekly stall; starting with it removes the whole class.
DECISION	proposed

PM-004-R2 P1 **Have the pacer run the testbed bring-up script on resume so containers that vanish during a pause come back before work starts.**

TARGET	`project`
BENEFIT / EFFORT	2/5 · 1/5 → score 6
RISK	Bring-up on every resume adds a minute.
ALTERNATIVES	Restart policies on the containers; manual recovery as now.
EVALUATION	One script call covers every pause cause, not only usage pauses.
DECISION	proposed

Tooling & automation

PM-013 Tooling and paths lagged the code moves CONFIRMED severity 2/5 · confidence 80%

OBSERVATION

The restructure tooling lagged the moves it served: movable.py was extended step after step as gates found misses (fully qualified names, sealed types, comments in strings); path-bearing docs and CI checks drifted after moves at least three times; Node was missing so gen-tokens.mjs was edited but never run, and a token-generator path bug reached a pushed state.

IMPACT

Repeated gate failures and fixes; one escaped bug found by review after the Phase 1 close.

EVIDENCE

1. FILE DEVLOG.md:259,284,476 — movable.py missed cases found by gates
2. FILE DEVLOG.md:357 — post-close review: generator would have produced duplicate classes

ROOT CAUSE

Generators and path checks were not part of the gate, and a needed tool (Node) was not installed, so drift was found late.

> **AUDIT · REVIEWER SUBAGENT (SONNET) (INDEPENDENT)**
>
> **Method:** Read D:212, 257-259, 284, 343-357, 476; git log shows d96d46b99 (generated tokens moved to shared) after 0f5c03482 (Phase 1 close). Dropped D:328 from the miss list.
>
> **Counter-evidence sought:** D:328 is a planned capability extension, not a gate miss; D:212 says no CI script referenced moved files, so CI drift is D:258 and D:357 only. 'At least three' drift events still holds across docs and CI; Node missing confirmed at D:257 and D:357.

PM-013-R1 P3 **Run every generator in the gate (install Node for gen-tokens) and add a check that every path named in docs, scripts and CI exists.**

TARGET	tool:path-check
BENEFIT / EFFORT	2/5 · 2/5 → score 1.6
RISK	Gate time grows slightly.
ALTERNATIVES	Fix drift as reviews find it.
EVALUATION	Turns a recurring late finding into an immediate one.
DECISION	proposed

Environment & lab

PM-003 **First execution session could not build anything** CONFIRMED severity 3/5 · confidence 85%

OBSERVATION

The first execution session ran in a cloud container whose proxy blocked Maven Central, Google Maven, Gradle, the JetBrains Compose repo, Flathub and the GitHub Actions API. The block was confirmed within about two minutes of the working request, before any spike code; no spike could be built, so the session produced a code survey, plan corrections, a proposal draft and a handoff, and the owner moved development to Claude Code on their own machine.

IMPACT

One session (about an hour) spent without builds; Phase 0 spikes moved to a second session.

EVIDENCE

1. FILE `DEVLOG.md:19-50` — Session 1: hosts returned 403; owner chose Claude Code on own computer
2. TRANSCRIPT `planning session, proxy status check` — recentRelayFailures: connect_rejected for repo.maven.apache.org, dl.google.com, plugins.gradle.org, dl.flathub.org

ROOT CAUSE

The execution surface was chosen before checking that it could reach the package hosts an Android/Kotlin/Flatpak build needs.

> AUDIT · REVIEWER SUBAGENT (OPUS) (INDEPENDENT)
>
> **Method:** Read DEVLOG 19-50, transcript idx 172-188 (curl 000 for Maven, Google, JetBrains, Gradle, Flathub; proxy 403 CONNECT failures; Actions API 403) and idx 263-264 (owner chose Claude Code), and the phase-0 handoff. Corrected the timing.
>
> **Counter-evidence sought:** No spike code preceded the check (idx 156-164 only create tasks). The session also corrected plan facts and drafted the upstream proposal, nothing committed. 'About 5 minutes' was wrong: the block was confirmed about 1.5 minutes after the working prompt (7.5 after the first, interrupted one).

PM-003-R1 P1 **Run build-heavy projects in Claude Code on the owner's machine (or the VM lab) from the first execution session, and keep claude.ai/Cowork for planning and reviews.**

TARGET	workflow
BENEFIT / EFFORT	3/5 · 1/5 → score 7.7
RISK	Planning and execution split across surfaces needs a clean handoff (this project's CLAUDE.md shows it works).
ALTERNATIVES	Request allowlist changes for the cloud proxy; keep using the cloud and push to CI for builds.
EVALUATION	The owner's machine already has the toolchain, network and devices; allowlist changes are outside the owner's control.
DECISION	proposed

Communication & handoffs

PM-001 **First plan borrowed another project's design** CONFIRMED severity 3/5 · confidence 90%

OBSERVATION

In the claude.ai planning session the first Argosy Desktop plan used an HTML/CSS/JS web-theme UI in an Electron shell, the defining design of the owner's separate portable-Argosy project (later named FreightCar) held in memory, whose alias 'argosy port' matched the request's wording, and it also carried that project's 'no integrated emulators' scope. It did not ask or offer porting Argosy's own UI. The owner had to redirect it as a separate, faithful port, and the plan was rewritten around Compose Multiplatform.

IMPACT

One full plan draft discarded and rewritten (about an hour of the planning session); risk of the wrong architecture reaching execution.

EVIDENCE

1. TRANSCRIPT `planning session, Docs calls idx 14-34` — first plan sections: 'web theme UI', 'Electron', 'Tauri'
2. USER `planning session idx 44, 2026-10-16 08:42 UTC` — 'Treat this chat as a completely separate project … wouldn't be able to tell the difference'
3. TRANSCRIPT `planning session, Docs calls idx 51-79` — rewrite around Compose Multiplatform

ROOT CAUSE

The assistant matched the request to a remembered project through an overlapping alias, treated that project's architecture and scope as context, and did not state the assumption or ask before drafting.

> AUDIT · REVIEWER SUBAGENT (OPUS) (INDEPENDENT)
>
> **Method:** Parsed the planning transcript: request at idx 1, memory read of the portable-Argosy file at idx 2, first-plan sections at idx 14/18/28 ('web theme UI', 'Electron', 'Why Electron over Tauri'), owner redirect at idx 44, rewrite from idx 45. Re-scoped: the project was not yet named FreightCar; the match came through the alias 'argosy port'.
>
> **Counter-evidence sought:** Checked for any question to the owner between idx 2 and 42 (none; only memory, search, fetch and Docs calls) and for any offer to port Argosy's own Compose UI in the first plan (none). The alias overlap explains the mix-up but does not remove it; the first plan also carried that project's no-integrated-emulators scope.

PM-001-R1 P1 **When a request matches more than one remembered project, name the project and the scope the plan assumes in the first reply, and ask if the request does not settle it, before drafting.**

TARGET	`workflow`
BENEFIT / EFFORT	3/5 · 1/5 → score 8.1
RISK	One extra question at project start; can feel redundant when the request is clear.
ALTERNATIVES	Do nothing and rely on the owner to correct; drop memory from planning sessions entirely.
EVALUATION	Cheapest way to stop cross-project contamination while keeping memory's benefits; dropping memory loses useful context.
DECISION	proposed

PM-001-R2 P1 **Open new projects that sit next to an existing one with a one-line scope and non-goal statement (for example 'separate from FreightCar; faithful port').**

TARGET	`user`
BENEFIT / EFFORT	2/5 · 1/5 → score 5.4
RISK	Small habit cost for the owner.
ALTERNATIVES	Rely on R1 alone.
EVALUATION	Complements R1 at near-zero cost; the owner's later instruction fixed the plan in one message.
DECISION	proposed

PM-002 Execution started before the workspace was known PLAUSIBLE severity 2/5 · confidence 62%

OBSERVATION

When the owner first asked to execute the plan, the assistant's first shell call cloned upstream into the cloud container, without knowing where development should live. The owner interrupted it (at the moment the owner's computer was being linked) and re-sent the request about six minutes later with the local workspace path, the GitHub remote, the README rule and the dev-log rule.

IMPACT

About 6 minutes and one owner interruption; no harm done.

EVIDENCE

1. SIGNAL `S-003` — user interrupted 1x
2. TRANSCRIPT `planning session idx 91-97` — Bash git clone refused; request re-sent with workspace and repo

ROOT CAUSE

The assistant began executing without confirming where the project lives and how it must be recorded; those requirements were only in the owner's head at that point.

> **AUDIT · REVIEWER SUBAGENT (SONNET) (INDEPENDENT)**
>
> **Method:** Read planning transcript lines 1-97: request at 10:10 PT (88), memory read and ToolSearch (89), git clone in the cloud container (91), refusal and interrupt (92-93), re-sent request at 10:16 (97); grepped for a workspace path before line 86. Re-scoped: interrupted at the moment the owner's computer was linked, rather than 'refused'.
>
> **Counter-evidence sought:** No workspace path appears before line 86, so the location was genuinely unknown. But the interrupt landed within seconds of the 'session linked to your computer' reminder, the transcript does not say why the owner interrupted, and line 94 is a harness rewind entry; the causal link is inferred from the re-sent content.

PM-002-R1 P2 **Before executing a plan that creates or changes code, confirm the workspace path, remote and record-keeping rules in the first reply unless a project file already states them.**

TARGET	workflow
BENEFIT / EFFORT	2/5 · 1/5 → score 2.5
RISK	Adds one round trip when the owner is away; mitigated by stating a default and proceeding when unattended.
ALTERNATIVES	Start in a scratch location and move later; do nothing.
EVALUATION	A one-line confirmation costs less than an interrupted start and avoids work in the wrong place.
DECISION	proposed

PM-009 Owner-dependent items stall Phase 5 PLAUSIBLE severity 3/5 · confidence 70%

OBSERVATION

Since 2026-11-05 the pacer is 'waiting-owner' on four items: Deck Game Mode checks, RetroArch cores, app id and screenshots, Brightness. They were first raised 7-12 days earlier (Deck checks 10-24, Brightness 10-26, app id 10-27, cores 10-29). The handoff says nothing is left that does not need the owner. There were no local commits from 10-30 to 11-04 and none after 11-05 01:39, apart from owner-requested features on 11-04/05. Daily updates go to the Claude Code session, not the owner's phone.

IMPACT

Plan steps idle for most of a week; step 4 and the Flathub submission blocked.

EVIDENCE

1. FILE `.git/argosy-pacer.json` — waiting-owner, 2026-11-05, four items
2. FILE `docs/desktop/handoff/phase-5.md:58` — nothing left that does not need the owner
3. LOG `git reflog` — 127 h gap 10-30 06:54 → 11-04 13:37; last commit 11-05 01:39

ROOT CAUSE

Owner questions are recorded in files the owner does not routinely read; nothing actively alerts the owner when work blocks on them.

> **AUDIT · REVIEWER SUBAGENT (SONNET) (INDEPENDENT)**
>
> **Method:** Read pacer JSON, handoff H:22/41/45/58, D:1327/1349/1364/1412-1416/1562, reflog.tsv, CLAUDE.md lines 23-24. Corrected ages to 7-12 days and noted owner-requested work on 11-04/05.
>
> **Counter-evidence sought:** First-raised dates hold (Deck 10-24, Brightness 10-26, app id 10-27, cores 10-29) and the 126.7 h commit gap holds, but '10-11 days' is right only for two items; the owner was active on 11-04/05 with new requests, the 10-28 pause was owner-directed, and the waiting-owner stamp is one day old. No log entry explains the 10-30 to 11-04 gap.

PM-009-R1 P2 **When the pacer enters waiting-owner, send a phone push notification listing each owner item and how long it has waited.**

TARGET	`tool:pacer-notify`
BENEFIT / EFFORT	4/5 · 2/5 → score 4.2
RISK	Notification fatigue if items are trivial.
ALTERNATIVES	Daily update only (as now); email digest.
EVALUATION	The owner asked for push notifications when a project needs intervention; it targets the exact blocking moment.
DECISION	proposed

PM-009-R2 P1 **Keep an owner-queue table (item, raised on, what it blocks) at the top of the current handoff, and ask each item when it arises instead of at step end.**

TARGET	`project`
BENEFIT / EFFORT	3/5 · 1/5 → score 6.3
RISK	Slightly longer handoff.
ALTERNATIVES	Leave questions in DEVLOG prose.
EVALUATION	Makes the waiting time visible and lets the owner answer in one pass.
DECISION	proposed

PM-016 Append-only log makes old status read as current PLAUSIBLE severity 1/5 · confidence 68%

OBSERVATION

The append-only DEVLOG makes older status lines read as current unless a reader reads on: 'Step 4 closed. Alpha republished' (10-26) is only reversed in a later entry, and headings keep the step numbers in force when written. gotchas.md's header still says it was collected from Phase 1 and 2 handoffs, and DEVLOG has grown to ~410 KB in one file. The handoff itself is current.

IMPACT

Risk of a fresh session acting on stale status; harder for the owner to follow progress.

EVIDENCE

1. FILE `DEVLOG.md:1351` — 'Step 4 closed. Alpha republished'
2. FILE `docs/desktop/handoff/phase-5.md:19` — Step 4: in progress
3. FILE `DEVLOG.md:1404-1447,1499` — headings keep old step numbers after renumbering

ROOT CAUSE

DEVLOG is append-only by design, so status changes (reopen, renumber) are recorded later rather than corrected where a reader looks first.

> **AUDIT · REVIEWER SUBAGENT (SONNET) (INDEPENDENT)**
>
> **Method:** Read D:1351, D:1404, D:1447, D:1495-1500, H:5/19/34/36, G:3, G:58 and ran wc -c on DEVLOG (409,644 bytes, 1,634 lines, 49 lines over 1,000 characters). Re-scoped to severity 1.
>
> **Counter-evidence sought:** DEVLOG is an append-only dated log: 'Step 4 closed' (10-26) was true when written and the 10-27 entry explicitly reopens and renumbers; the handoff is current. What remains is that a reader of an older entry must read on to learn its status, the gotchas header is stale, and the log is one 410 KB file.

PM-016-R1 P3 **Keep a step-status table at the top of each phase section in DEVLOG.md, updated whenever a step closes, reopens or is renumbered.**

TARGET	project
BENEFIT / EFFORT	2/5 · 1/5 → score 1.4
RISK	One more thing to maintain per unit.
ALTERNATIVES	Split DEVLOG per phase; rely on the handoff only.
EVALUATION	Keeps the owner's single-document log and gives a correct status at a glance.
DECISION	proposed

PM-019 Per-step approvals until the owner removed them CONFIRMED severity 2/5 · confidence 85%

OBSERVATION

In Phase 1 the owner answered 'proceed' after several already-approved steps (five or more round trips) until the owner told Claude to stop asking between steps and ask only at phase boundaries.

IMPACT

Five or more owner round trips; work idle while waiting for each.

EVIDENCE

1. FILE DEVLOG.md:191,206,219,232,246,267 — 'Owner said proceed' for steps 3-7 and the emulator setup
2. FILE DEVLOG.md:312 — 'stop asking to proceed between steps; ask only at major phase boundaries'

ROOT CAUSE

The approval granularity was not stated at the start; the assistant defaulted to asking at every step.

> **AUDIT · REVIEWER SUBAGENT (SONNET) (INDEPENDENT)**
>
> **Method:** Read D:191, 206, 219, 224, 232, 246, 267, 312 and surrounding headings. Corrected the evidence lines.
>
> **Counter-evidence sought:** 'Owner said proceed' appears for steps 3-7 (D:206, 219, 232, 246, 267) plus D:191; D:224 is a testbed note, not a proceed. D:312 follows the last round trip. Whether every step was already approved was not checked in the planning transcript.

PM-019-R1 P2 State the approval granularity (phase gates only, plus destructive or third-party actions) in each new project's CLAUDE.md from day one.

TARGET	workflow
BENEFIT / EFFORT	2/5 · 1/5 → score 3.4
RISK	Fewer check-ins could let a wrong direction run longer.
ALTERNATIVES	Ask at every step (as at first).
EVALUATION	Phase gates plus the destructive-action rule kept control here without the round trips.
DECISION	proposed

Quality & testing

PM-007 **Test walks ran on the owner's real desktop and data** CONFIRMED severity 4/5 · confidence 80%

OBSERVATION

Automated desktop walks from Phase 2 on ran in the owner's live session, and in Phases 3-4 this had side effects. A walk pressed 'Scan for Android games', adding 13 of the owner's desktop entries to the library and sending the owner's app ids to Google's Play Store lookup; another toggled a real setting; a sign-in opened the owner's browser; a full-screen capture grabbed other monitors. Each was disclosed and the Play Store path was gated; a throwaway-data-dir walk existed from Phase 3 step 9 but was not used consistently.

IMPACT

Owner's installed-app list sent to a third party; owner data changed and needing cleanup; risk to the owner's real setup until the Phase 5 VM lab.

EVIDENCE

1. FILE `DEVLOG.md:989-995` — scan added rows 18-30; 'sends the owner's app ids to Google'; PlayStoreAccess gate
2. FILE `DEVLOG.md:1253` — a walk toggled 'Close Emulator When Returning to Argosy' in the owner's real data
3. FILE `DEVLOG.md:852,1012` — full-screen capture of other monitors; pair URL opened in host browser

ROOT CAUSE

No isolated desktop test environment existed until the Phase 5 VM lab, and the partial isolation (a throwaway data dir) was not the default; walks used the owner's $HOME, desktop entries, browser and network.

> AUDIT · REVIEWER SUBAGENT (OPUS) (INDEPENDENT)
>
> **Method:** Read DEVLOG 850-852, 986-997, 1006-1040, 1010-1012, 1253, 1478-1494, phase headers, and Phase 2 walk entries 662-741. Re-scoped: incidents in Phases 3-4; walks from Phase 2 used the owner's session.
>
> **Counter-evidence sought:** All four incidents confirmed with the Play Store 404s logged (D:990) and the gate fix (D:994-995); spectacle capture is D:852. Partial isolation existed from Phase 3 step 9 (fresh-walk.sh with a throwaway XDG_DATA_HOME) but was not applied consistently: the Phase 4 settings slip came after it. The 13 rows were left for an owner answer, not cleaned up. Phase 2 walks had no recorded side effects.

PM-007-R1 P1 **Run every GUI walk in the VM lab or a disposable user with separate HOME/XDG dirs and testbed-only network, and make the walk scripts refuse to start against the owner's real HOME.**

TARGET	`project`
BENEFIT / EFFORT	5/5 · 2/5 → score 8
RISK	Walks get slower and VM-specific failures appear (Phase 5 shows the cost).
ALTERNATIVES	Keep host walks with a stricter checklist; mock the host shell.
EVALUATION	Only isolation removes the class (privacy leak and data changes); a checklist depends on care.
DECISION	proposed

PM-007-R2 P1 **Add 'GUI tests never run against the owner's live session or data; scripts must check' as a required rule in the VM Lab Standard.**

TARGET	`standard:VM-LAB-STANDARD`
BENEFIT / EFFORT	4/5 · 1/5 → score 12.8
RISK	Projects without a lab need a disposable-user fallback.
ALTERNATIVES	Leave it to each project's CLAUDE.md.
EVALUATION	A standard rule applies to every future project at the cost of one paragraph.
DECISION	proposed

PM-008 Target-hardware checks lag; the Game Mode fix is unverified PLAUSIBLE

severity 4/5 · confidence 60%

OBSERVATION

Verification ran on the testbed, the owner's desktop with virtual pads, and later VMs. The first run on a separate target device with a physical pad was the owner's Bazzite HTPC test on 10-23 (15 pass, 3 fail, 13 skipped). A Game Mode launch fix shipped as alpha 5.4 on code-reading, jshell and ldd evidence; step 4 was declared closed on 10-26 and reopened on 10-27, and the fix has still not been verified in real Game Mode.

IMPACT

Phase 5's central goal is unverified; the step was closed and reopened; controller and Game Mode issues surface only when the owner tests.

EVIDENCE

1. FILE DEVLOG.md:1100-1102 — Bazzite owner test: 15 pass, 3 fail, 13 skipped
2. FILE DEVLOG.md:1335-1351 — Game Mode fix on code-reading; nested-gamescope test blocked; 'Step 4 closed'
3. FILE .git/argosy-pacer.json — waiting-owner: Deck Game Mode checks

ROOT CAUSE

Target hardware (Steam Deck, Steam Controller, AMD/Intel GPUs) is outside the agent's reach, and owner hardware checks were batched at phase ends rather than tied to each step.

> **AUDIT · REVIEWER SUBAGENT (OPUS) (INDEPENDENT)**
>
> **Method:** Read DEVLOG 796-798, 852, 1003, 1035, 1100-1102, 1301-1351, 1493-1502, phase-5 handoff 1-45, pacer json, reflog 10-24 to 10-27. Re-scoped to: first separate target device and physical pad on 10-23; Game Mode fix shipped and step closed without on-hardware verification.
>
> **Counter-evidence sought:** Many walks ran on the owner's physical desktop (real RetroArch launch D:852, wpctl volume D:1003, KDE portal D:1035), so 'almost entirely testbed' overstates it; what was missing was a separate target device and a physical pad (D:798). D:1321 and D:1335 suggest an earlier owner Game Mode run found the failures, so 'never run' is doubtful; what is certain is that the fix has not been verified in real Game Mode. Bazzite 15/3/13, step close 10-26 and reopen 10-27, and the pending Deck checks are confirmed.

PM-008-R1 P2 **Attach a 3-5 item real-hardware checklist to every alpha publish, and do not mark a hardware-dependent step closed on testbed evidence alone.**

TARGET	workflow
BENEFIT / EFFORT	4/5 · 2/5 → score 4.8
RISK	More frequent owner involvement; steps wait on the owner.
ALTERNATIVES	Batch owner tests at phase ends (as now); rely on VMs.
EVALUATION	Small, frequent checks catch hardware issues while the code is fresh; VMs cannot run Game Mode.
DECISION	proposed

PM-008-R2 P2 **Make the Steam Deck reachable as a scripted test device (SSH from the dev machine with a launch-and-log script) so Game Mode runs need the owner only to switch modes.**

TARGET	project
BENEFIT / EFFORT	4/5 · 3/5 → score 3.2
RISK	Setup effort and device access to manage.
ALTERNATIVES	Owner runs every Deck check by hand.
EVALUATION	Turns the Phase 5 blocker into a repeatable check.
DECISION	proposed

PM-012 Checks confirmed outcomes, not the specific effect PLAUSIBLE severity 2/5 · confidence 70%

OBSERVATION

Two behaviours were recorded as verified and later found wanting: the Phase 5 exit cleanup was dead code because Compose's application{} exits the process (claimed 10-24, found 11-04) while the check 'the run exits on its own' passed; and the Close Argosy row, reached by gamepad in its check, showed only as a sliver at the end of Quick Settings, found 11 days later while investigating the owner's 'missing exit button' report.

IMPACT

Two defects shipped in alphas and were reworked 11 days later; a later shutdown hook limited the practical effect of the first.

EVIDENCE

1. FILE DEVLOG.md:1321-1322,1625 — exit cleanup claimed; exitProcessOnExit made it dead code
2. FILE DEVLOG.md:1604-1606 — owner could not find the exit row; only a sliver showed
3. FILE DEVLOG.md:1345 — later shutdown hook releases SDL on close

ROOT CAUSE

Verification checked that the overall flow ended as expected rather than observing the claimed effect itself (the cleanup ran; the element is visible at the default window size).

> **AUDIT · REVIEWER SUBAGENT (OPUS) (INDEPENDENT)**
>
> **Method:** Read DEVLOG 1321, 1345, 1604-1606, 1625 and 431 with reflog times. Split the claim: (a) exit cleanup confirmed (85), (b) Close row partly holds (60), (c) ThemeViewModel crash rejected as an instance (moved to PM-021). Severity lowered to 2.
>
> **Counter-evidence sought:** (a) holds: the check was 'the run exits on its own', and exitProcessOnExit made the cleanup dead for 11 days, though a shutdown hook added 10-26 limited the effect. (b) the row worked by gamepad (DOWN*8) and the sliver was found by the assistant while investigating the owner's report, which concerned the drawer. (c) was caught in the same work unit by the device run before being logged as done.

PM-012-R1 P2 **For each claimed behaviour, verify its own observable effect (a log line, file, row or a screenshot of that element in the default window size) and name that check in the DEVLOG entry.**

TARGET	claude-md
BENEFIT / EFFORT	3/5 · 1/5 → score 4.2
RISK	Slightly longer verification per step.
ALTERNATIVES	Rely on the owner's tests to catch misses.
EVALUATION	Targets the exact gap in both cases at almost no cost.
DECISION	proposed

Standards & ai-assets

PM-015 Process rules changed often, and tooling followed PLAUSIBLE severity 2/5 · confidence 65%

OBSERVATION

Rules changed within days and the tooling had to follow: alpha publishing went from 'only at a phase close' to 'after every closed step' about 5 hours later, with push-pages.sh flags reworked; session clearing at unit end was added on 10-17 and removed on 10-23; early publishes were labelled 'step 5.4 close' while step 4 was open. Each publish is a ~240 MB single-commit force-push taking about 7 minutes.

IMPACT

Tooling rework, mislabelled publishes, and ~7 minutes plus 240 MB upload per step close.

EVIDENCE

1. FILE DEVLOG.md:1164-1169,1213-1217 — publish rule set then reversed
2. FILE DEVLOG.md:1151-1154 — no session clearing at unit end
3. FILE DEVLOG.md:1610 — early publish labelled 'step 5.4 close'

ROOT CAUSE

Rules were adopted without stating their recurring cost or tooling impact up front, and the publish script had no truthful label for an early publish.

> **AUDIT · REVIEWER SUBAGENT (SONNET) (INDEPENDENT)**
>
> **Method:** Read D:504-508, 1068, 1151-1154, 1164-1169, 1213-1217, 1610, H:36, G:58 and reflog commit times. Corrected the rule-reversal interval to about 5 hours.
>
> **Counter-evidence sought:** Session clearing added 10-17 and removed 10-23, the publish flags reworked, the 'step 5.4 close' label, ~240 MB and ~7 min all hold. The reflog shows the phase-close-only rule at 10-23 20:14 and the per-step rule at 10-24 01:01, about 5 h apart, not 12 h. Both changes were owner decisions applied promptly: process churn, not a defect.

PM-015-R1 P2 **When the owner proposes a process rule, state its recurring cost (time, upload, usage) and any tooling change before adopting it, and keep all rules in one dated CLAUDE.md table.**

TARGET	workflow
BENEFIT / EFFORT	2/5 · 1/5 → score 2.6
RISK	Slight friction when the owner wants a quick change.
ALTERNATIVES	Adopt rules as given (as now).
EVALUATION	The owner decides with the cost in view; reversals become rarer.
DECISION	proposed

PM-015-R2 P3 **Publish gh-pages incrementally instead of a single-commit force-push, and add an --early flag so early publishes are labelled truthfully.**

TARGET	project
BENEFIT / EFFORT	2/5 · 2/5 → score 1.3
RISK	Repository history grows; needs occasional pruning.
ALTERNATIVES	Keep force-pushing.
EVALUATION	Cuts upload time per step and fixes the mislabelling.
DECISION	proposed

PM-017 Hooks and permission checks blocked routine work CONFIRMED severity 2/5 · confidence 80%

OBSERVATION

Upstream's Claude hooks and the session permission layer repeatedly got in the way: in Phase 0 every Bash call was refused because the hooks' project dir pointed at an old folder, needing a new session; the verification-guard hook's push detection (meant for pushes carrying Kotlin without fresh tests) fires on 'git push' text even inside a heredoc, so push scripts are written to files and run by path; a nested-gamescope test was blocked by the permission classifier.

IMPACT

One session restart in Phase 0; workarounds for every push; one Game Mode test not run.

EVIDENCE

1. FILE DEVLOG.md:74 — every Bash call refused; stale project dir
2. FILE DEVLOG.md:1071 — hook blocks 'git push' text even in a heredoc
3. FILE DEVLOG.md:1349 — nested-gamescope test blocked by the permission classifier

ROOT CAUSE

Upstream's hooks were written for upstream contributors' workflow, and the desktop port's routine operations (pushing at phase close, process tests) were never allowed for explicitly.

> **AUDIT · REVIEWER SUBAGENT (SONNET) (INDEPENDENT)**
>
> **Method:** Read D:74, D:1071, D:1349, G:58 and verification-guard.py lines 25-140. Narrowed the push-hook wording.
>
> **Counter-evidence sought:** The hook only blocks pushes carrying Kotlin without fresh unit-test results and has bypasses (ARGOSY_SKIP_VERIFY_CHECK=1, --dry-run), so 'blocks any git push text' is overbroad; its text match firing inside a heredoc rests on D:1071 and G:58 (the git_command module was not found to confirm). The Phase 0 stale-project-dir block and the classifier block are as stated.

PM-017-R1 P2 **Add a desktop-port allowance for scripts/desktop/** to the verification-guard hook (or one documented push wrapper), and add a narrow permission rule for the gamescope test command.**

TARGET	project
BENEFIT / EFFORT	2/5 · 1/5 → score 3.2
RISK	Loosens a guard upstream relies on; keep the change desktop-scoped.
ALTERNATIVES	Keep working around it.
EVALUATION	A scoped allowance keeps upstream's protection while removing the recurring workaround.
DECISION	proposed

PM-018 Post-mortem evidence had to be reconstructed CONFIRMED severity 2/5 · confidence 85%

OBSERVATION

The post-mortem's evidence base is incomplete: there is no session journal, only the claude.ai planning transcript was available, because the Claude Code transcripts on the owner's computer are unreachable from the surface that ran this review, and the plan lives in PORT_PLAN.md, which the miner does not read (it only parses docs/PLAN.md). Plan progress and sessions were reconstructed from DEVLOG, handoffs and the git reflog.

IMPACT

No token, tool-call or interrupt metrics for the Claude Code sessions; weaker evidence for usage findings.

EVIDENCE

1. SIGNAL `S-002` — JOURNAL.md missing
2. METRIC `metrics.json` — plan: null; 1 transcript session (planning only)
3. FILE `scripts/postmortem.mjs parsePlan` — reads docs/PLAN.md only

ROOT CAUSE

The Post-Mortem Standard was adopted after the project began, and its tooling assumes a fixed plan path and same-surface transcripts.

> **AUDIT · REVIEWER SUBAGENT (SONNET) (INDEPENDENT)**
>
> **Method:** Read metrics.json (plan and journal null, S-002 no-journal, one session), postmortem.mjs parsePlan (docs/PLAN.md only), listed docs/ and repo root (PORT_PLAN.md at root, no JOURNAL.md).
>
> **Counter-evidence sought:** Transcripts are not wholly missing: the 10-16 claude.ai planning transcript was mined; only the Claude Code transcripts on the owner's computer were unreachable, which could not be checked from here.

PM-018-R1 P1 Start docs/postmortem/JOURNAL.md now and add one line per session on every surface for the rest of Phase 5.

TARGET	`project`
BENEFIT / EFFORT	3/5 · 1/5 → score 5.1
RISK	Small per-session cost.
ALTERNATIVES	Rely on DEVLOG alone.
EVALUATION	DEVLOG records the work but not issues and corrections per session, which the standard needs.
DECISION	proposed

PM-018-R2 P2 Let postmortem.mjs take --plan <file> and accept 'Phase N · Title' headings.

TARGET	`standard:POST-MORTEM-STANDARD`
BENEFIT / EFFORT	2/5 · 1/5 → score 3.4
RISK	Script change needs a version bump and re-copy into projects.
ALTERNATIVES	Rename plans to docs/PLAN.md.
EVALUATION	Projects keep their own plan names and still get plan metrics.
DECISION	proposed

PM-018-R3 P1 **Run final post-mortems from Claude Code on the owner's machine so its transcripts are mined.**

TARGET	workflow
BENEFIT / EFFORT	3/5 · 1/5 → score 5.1
RISK	Needs a Claude Code session at the end.
ALTERNATIVES	Copy transcripts to the project folder for Cowork.
EVALUATION	Simplest way to get full metrics; copying transcripts risks leaking them into the repo.
DECISION	proposed

5. Recommendation register

All recommendations from confirmed and plausible findings, ranked by score.

PRI	ID	ACTION	TARGET	SCORE	DECISION
P1	PM-010-R2	Decide on the upstream proposal now: post it, or record 'long-term fork' and plan for the merge cost.	user	14.1	proposed
P1	PM-007-R2	Add 'GUI tests never run against the owner's live session or data; scripts must check' as a required rule in the VM Lab Standard.	standard:VM-LAB-STANDARD	12.8	proposed
P1	PM-010-R1	Merge each upstream stable release into main within a week using reapply.sh (at least at every phase gate), and add 'upstream releases behind' to the daily update.	claude-md	8.8	proposed
P1	PM-001-R1	When a request matches more than one remembered project, name the project and the scope the plan assumes in the first reply, and ask if the request does not settle it, before drafting.	workflow	8.1	proposed
P1	PM-007-R1	Run every GUI walk in the VM lab or a disposable user with separate HOME/XDG dirs and testbed-only network, and make the walk scripts refuse to start against the owner's real HOME.	project	8	proposed
P1	PM-003-R1	Run build-heavy projects in Claude Code on the owner's machine (or the VM lab) from the first execution session, and keep claude.ai/Cowork for planning and reviews.	workflow	7.7	proposed
P1	PM-009-R2	Keep an owner-queue table (item, raised on, what it blocks) at the top of the current handoff, and ask each item when it arises instead of at step end.	project	6.3	proposed
P1	PM-004-R1	Make work units, a rewritten per-phase handoff, quiet build output and usage pacing the default bootstrap of every multi-day project (fold this project's scheme into the Token Usage Standard).	standard:TOKEN-USAGE-STANDARD	6	proposed
P1	PM-004-R2	Have the pacer run the testbed bring-up script on resume so containers that vanish during a pause come back before work starts.	project	6	proposed
P1	PM-001-R2	Open new projects that sit next to an existing one with a one-line scope and non-goal statement (for example 'separate from FreightCar; faithful port').	user	5.4	proposed
P1	PM-018-R1	Start docs/postmortem/JOURNAL.md now and add one line per session on every surface for the rest of Phase 5.	project	5.1	proposed
P1	PM-018-R3	Run final post-mortems from Claude Code on the owner's machine so its transcripts are mined.	workflow	5.1	proposed
P2	PM-008-R1	Attach a 3-5 item real-hardware checklist to every alpha publish, and do not mark a hardware-dependent step closed on testbed evidence alone.	workflow	4.8	proposed
P2	PM-009-R1	When the pacer enters waiting-owner, send a phone push notification listing each owner item and how long it has waited.	tool:pacer-notify	4.2	proposed
P2	PM-012-R1	For each claimed behaviour, verify its own observable effect (a log line, file, row or a screenshot of that element in the default window size) and name that check in the DEVLOG entry.	claude-md	4.2	proposed
P2	PM-005-R1	Replace the week-based roadmap with actual dates plus remaining work in units, and re-baseline at every phase gate, naming the binding constraint (usage window or owner action).	project	3.4	proposed
P2	PM-018-R2	Let postmortem.mjs take --plan <file> and accept 'Phase N · Title' headings.	standard:POST-MORTEM-STANDARD	3.4	proposed

P2	PM-019-R1	State the approval granularity (phase gates only, plus destructive or third-party actions) in each new project's CLAUDE.md from day one.	workflow	3.4	proposed
P2	PM-006-R1	At plan approval, walk the parity matrix row by row with the owner as keep / remove / later, and put a clickable build in front of the owner by the end of the first build phase.	workflow	3.2	proposed
P2	PM-008-R2	Make the Steam Deck reachable as a scripted test device (SSH from the dev machine with a launch-and-log script) so Game Mode runs need the owner only to switch modes.	project	3.2	proposed
P2	PM-011-R1	Add a gate check that fails on emptyFlow(), TODO() and constant false/null actuals in desktopMain unless listed in an allowlist with a reason.	tool:stub-guard	3.2	proposed
P2	PM-017-R1	Add a desktop-port allowance for scripts/desktop/** to the verification-guard hook (or one documented push wrapper), and add a narrow permission rule for the gamescope test command.	project	3.2	proposed
P2	PM-015-R1	When the owner proposes a process rule, state its recurring cost (time, upload, usage) and any tooling change before adopting it, and keep all rules in one dated CLAUDE.md table.	workflow	2.6	proposed
P2	PM-002-R1	Before executing a plan that creates or changes code, confirm the workspace path, remote and record-keeping rules in the first reply unless a project file already states them.	workflow	2.5	proposed
P3	PM-014-R2	Make save-proof.sh keep or restore the desktop's pairing so a proof never needs a separate re-pair.	project	1.8	proposed
P3	PM-013-R1	Run every generator in the gate (install Node for gen-tokens) and add a check that every path named in docs, scripts and CI exists.	tool:path-check	1.6	proposed
P3	PM-016-R1	Keep a step-status table at the top of each phase section in DEVLOG.md, updated whenever a step closes, reopens or is renumbered.	project	1.4	proposed
P3	PM-015-R2	Publish gh-pages incrementally instead of a single-commit force-push, and add an --early flag so early publishes are labelled truthfully.	project	1.3	proposed
P3	PM-014-R1	Build validated golden VM images per distro once (gamescope and testbed pairing baked in) and replace fixed sleeps with readiness markers from the app log.	standard:VM-LAB-STANDARD	0.9	proposed

Appendix A — Rejected observations

ID	OBSERVATION	WHY REJECTED	CONF.
PM-020	**Phase 3 started without asking the owner** Phase 2 was closed with exit criteria not fully met and Phase 3 was started without the owner being asked, contrary to the standing rule to stop and ask between phases.	The Phase 3 status line records 'owner: Start Phase 3 now.' and the 15:26 commit is 'Phase 3: start (owner go-ahead)', six minutes after the close: the owner was asked. Phase 2 did close with unmet items, but they were disclosed and carried forward; at most a severity-1 note.	80
PM-021	**A green gate let a DI crash through** Behaviours were recorded as verified and later found broken: the first device run after a fully green gate crashed with 'Cannot create an instance of class ThemeViewModel' because Hilt KSP did not see @HiltViewModel through expect/actual.	The crash was caught in the same work unit by the process's own device run, before the work was logged as done, and fixed with a generator task. It shows a gate blind spot that the device check covered, not a behaviour recorded as verified and later found broken.	75

Appendix B — Signal dispositions

SIGNAL	KIND	DETAIL	DISPOSITION
S-001	idle-gap	Idle 504 min mid-session (prompt cache expired) at 2026-10-16T08:46:43.470Z	expected: overnight gap between the plan revision (01:42 PT) and the owner's execute request (10:10 PT), not idle work
S-002	no-journal	docs/postmortem/JOURNAL.md missing: session history for non-Claude-Code work must be reconstructed	PM-018
S-003	interrupt	User interrupted 1× (possible wrong direction)	PM-002

Appendix C — Sessions

START	SESSION	MIN	TOKENS	CALLS	ERRORS	MODEL
2026-10-16 08:30	383d8d2b	536	12,324,275	106	4	claude-sonnet-4-6, claude-opus-5-5